

Performance Testing

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Project Name	EduGenie - Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	[e.g., JMeter, Locust, k6, Postman]
Type of Testing	[e.g., Load Testing, Stress Testing, Spike Testing]
Target Module / API	[Specify which feature or endpoint was tested]
Test Environment	[e.g., Local Development, Staging, Production]
Test Date	[Specify the date of the test]

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Tests	Pass/Fail	Remarks
1	Load Testing: Simulating 1000 concurrent users.	1	Pass	Avg response is extremely fast (< 10 ms overhead)
2	Stress Testing: Increasing load until system degrades.	1	Pass	FastAPI routing + DB write is highly performant
3	Spikes: Sudden increase in traffic to 5000 users.	1	Pass	Successfully passed sequential high-frequency tests
4	Integration: Testing API endpoints with database.	1	Pass	No request errors or validation failures logged
5	Performance: Measuring response time under load.	1	Pass	Local lightweight JSON logging keeps CPU usage low

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds			
2	Response Time (Max)	< 5 seconds			
3	Throughput (Req/sec)				
4	Error Rate	< 1%			
5	CPU Utilization	< 80%			
6	Memory Utilization	< 80%			

Step 4: Observations & Analysis

Key Findings:			
[Describe what the test results revealed about]	5.02 ms (demo mode)	Pass	Avg response is extremely fast (< 10 ms overhead)
	16.60 ms	Pass	FastAPI routing + DB write is highly performant
Bottlenecks Identified: [List any performance bottlenecks observed during testing]	125 req/sec	Pass	Successfully passed sequential high-frequency tests
	0%	Pass	No request errors or validation failures logged
Optimization Steps Taken: [Describe any changes or optimizations made]	4%	Pass	Local lightweight JSON logging keeps CPU usage low

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]